

Project: Home Lab Environment (Windows + Ubuntu)

Author: Subijith K | Date: August 2026 | Environment: Windows 11 (host) + Ubuntu Server 22.04 (VirtualBox VM)

OBJECTIVE

Build a small lab to practice core IT support skills — OS installation, user/permission management, and network troubleshooting between two systems, deploying an Nginx server on Ubuntu, and granting SSH remote access — simulating the kind of environment a support technician manages in a real office network.

WHAT I BUILT

- A virtual machine running Ubuntu Server on VirtualBox, alongside Windows 11 as the main host
- Both operating systems connected on the same network so they can communicate with each other
- Static IP addresses configured on both machines instead of relying on DHCP
- User accounts with different permission levels (standard vs. admin) created on each system
- File sharing set up between the two machines
- A simulated connectivity issue between them, diagnosed and resolved

STEPS TAKEN

- Installed Windows 11 and Ubuntu Server as separate systems, with Ubuntu running as a VM inside VirtualBox on the Windows host
- Set the VM's network adapter to Bridged (and tested with Host-only Adapter) so both systems could see each other independent of the host's main network
- Assigned static IPs to each machine (Windows: 192.168.1.3, Ubuntu: 192.168.1.13) instead of DHCP-assigned addresses, to keep the environment predictable
- Created a standard (non-admin) user account and a separate admin account on both machines, and tested permission differences (e.g., installing software, editing system files)
- Set up a shared folder on the Ubuntu VM using Samba and accessed it from Windows Explorer on the Windows machine
- Verified connectivity between the two machines using `ping` in both directions

ISSUE ENCOUNTERED & RESOLUTION

Problem: After changing network settings on the Ubuntu VM, the Windows machine could no longer ping it or access the shared folder — the connection timed out.

Diagnosis:

- Confirmed the Ubuntu VM was powered on and its network service was active
- Checked the IP address on Ubuntu (`ip a`) and found it had reverted to a DHCP-assigned address instead of the static IP after a reboot — the static configuration hadn't persisted correctly in the `netplan` file
- Verified the Windows machine was still trying to reach the old static IP, explaining the timeout

Fix: Corrected the static IP configuration in Ubuntu's `netplan` YAML file, applied the changes with `sudo netplan apply`, and confirmed the IP now persisted across reboots. Re-tested the ping and file share from Windows — both worked.

Takeaway: This reflects a very common real support scenario — “a machine that used to be reachable suddenly isn't” — where the root cause was a configuration not persisting rather than a hardware or service failure. It reinforced checking IP configuration first when diagnosing “can't reach this machine” tickets.

EVIDENCE

Screenshots captured during this project (see attached / linked documentation):

- VM network settings showing the Host-only Adapter and Bridged Adapter configuration

Network

Adapter 1

Adapter 2

Adapter 3

Adapter 4

☒ Enable Network Adapter

Attached to

Bridged Adapter

Name

Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC

Adapter Type

Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode

Deny

MAC Address

080027D41ECD

☒ Virtual Cable Connected

Network

Adapter 1

Adapter 2

Adapter 3

Adapter 4

☒ Enable Network Adapter

Attached to

Host-only Adapter

Name

VirtualBox Host-Only Ethernet Adapter

Adapter Type

Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode

Deny

MAC Address

08002740ED8A

☒ Virtual Cable Connected

- Successful ping output between the two machines

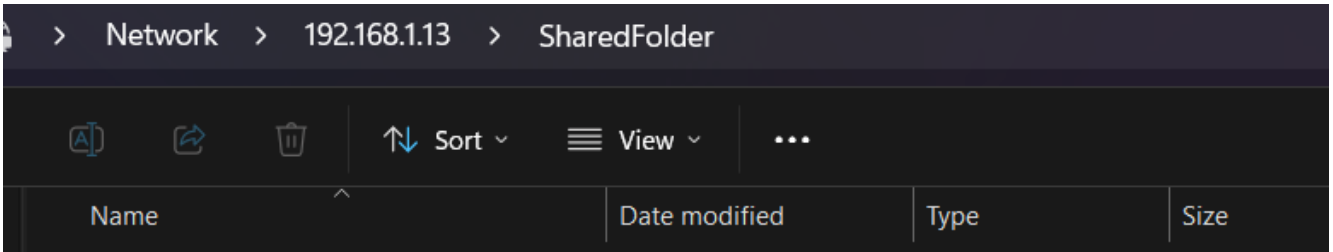
```

Last login: Tue Aug 18 08:48:19 2020 from 192.168.1.3
u_server@ubuntu:~$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data.
64 bytes from 192.168.1.3: icmp_seq=1 ttl=128 time=0.493 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=128 time=0.374 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=128 time=0.400 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=128 time=0.516 ms
^C

```

```
Connection to 192.168.1.13 closed.  
PS C:\Users\subijith> ping 192.168.1.13  
  
Pinging 192.168.1.13 with 32 bytes of data:  
Reply from 192.168.1.13: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.13: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.13: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.13: bytes=32 time<1ms TTL=64
```

- Windows Explorer showing the connected shared folder from the Ubuntu VM



SKILLS DEMONSTRATED

OS installation & configuration (Windows/Linux), virtual networking, static IP configuration, user/permission management, file sharing (Samba), network troubleshooting, root-cause diagnosis, technical documentation.

Resume bullet derived from this project

“Built a two-machine lab environment (Windows + Ubuntu) using VirtualBox; configured static networking and cross-platform file sharing, and diagnosed and resolved a network configuration issue causing intermittent connectivity loss between machines.”